

Problem 2

Evaluate the integral:

$$I = \int \frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1-x^4}} dx.$$

Solution

To solve this integral, we will simplify and manipulate the given expression step by step.

Step 1: Simplify the square roots

The denominator is given as $\sqrt{1-x^4}$. Notice that:

$$1 - x^4 = (1 - x^2)(1 + x^2).$$

Thus, we can rewrite the denominator as:

$$\sqrt{1-x^4} = \sqrt{(1-x^2)(1+x^2)} = \sqrt{1-x^2} \cdot \sqrt{1+x^2}.$$

Using this result, the integrand becomes:

$$\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1-x^4}} = \frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} \cdot \sqrt{1-x^2}}.$$

Separate the terms in the numerator:

$$\frac{\sqrt{1+x^2}}{\sqrt{1+x^2} \cdot \sqrt{1-x^2}} + \frac{\sqrt{1-x^2}}{\sqrt{1+x^2} \cdot \sqrt{1-x^2}}.$$

Simplify each term:

$$\frac{\sqrt{1+x^2}}{\sqrt{1+x^2} \cdot \sqrt{1-x^2}} = \frac{1}{\sqrt{1-x^2}}, \quad \frac{\sqrt{1-x^2}}{\sqrt{1+x^2} \cdot \sqrt{1-x^2}} = \frac{1}{\sqrt{1+x^2}}.$$

Thus, the integral becomes:

$$I = \int \left(\frac{1}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1+x^2}} \right) dx.$$

Step 2: Solve each term

The integral is now split into two simpler integrals:

$$I = \int \frac{1}{\sqrt{1-x^2}} dx + \int \frac{1}{\sqrt{1+x^2}} dx.$$

1. **First term: $\int \frac{1}{\sqrt{1-x^2}} dx$:** This is a standard integral:

$$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin(x).$$

2. **Second term: $\int \frac{1}{\sqrt{1+x^2}} dx$:** This is another standard integral:

$$\int \frac{1}{\sqrt{1+x^2}} dx = \operatorname{arsinh}(x) = \ln \left(x + \sqrt{x^2 + 1} \right).$$

Step 3: Combine the results

Combine both terms to get the final result:

$$I = \arcsin(x) + \ln\left(x + \sqrt{x^2 + 1}\right) + C,$$

where C is the constant of integration.

Final Answer

The solution to the integral is:

$$I = \arcsin(x) + \ln\left(x + \sqrt{x^2 + 1}\right) + C.$$